6 AI for Digital Public Safety: Defeating Counterfeiting, Fraud & Digital Arrest Scams

Theme: *Smart Cities / Public Safety / Digital Trust / Geospatial Law Enforcement*

PROBLEM CONTEXT

India registered 1.14 million cybercrime complaints in 2023 — up 60% from 2022 — and the trajectory has only steepened. The Ministry of Home Affairs reported that 'digital arrest' scams, where fraudsters impersonating CBI, ED, or Customs officers trap victims in multi-day psychological hostage situations over video call, defrauded citizens of over Rs 1,776 crore in just the first nine months of 2024. These are not opportunistic crimes — they are industrialised operations run from fraud compounds, often across borders, using spoofed numbers, AI-generated voices, and fake government portals. Separately, counterfeit currency remains a persistent threat: the RBI's Annual Report 2025 flagged record FICN (Fake Indian Currency Notes) seizures, with high-denomination Rs 500 fakes of sufficient quality to defeat manual detection in routine banking operations. What law enforcement lacks is not evidence after the fact — it is intelligence before mass victimisation occurs, and reliable tools to detect threats at the point of contact rather than the point of complaint. Solving this requires convergence of financial transaction intelligence, communication network analysis, physical security (counterfeit detection), and real-time public safety coordination — exactly the kind of multi-source, multi-agency intelligence problem where AI has the potential to be transformative.

CHALLENGE STATEMENT

Build an AI-powered Digital Public Safety Intelligence platform that equips law enforcement agencies, financial institutions, and citizens with proactive tools to detect, disrupt, and respond to digital fraud networks, counterfeit currency circulation, and organised scam operations — shifting from reactive case investigation to predictive threat neutralisation.

WHAT YOU MAY BUILD

Participants may explore areas such as:

- **Digital Arrest Scam Detection & Alerting —** Real-time AI classifier trained on digital arrest scam patterns — call flow sequences, number spoofing signatures, script templates, video call metadata — that flags active scam sessions to telecom providers and potential victims before financial transfer occurs, with automated MHA alert generation.
- **Counterfeit Currency Identification Agent —** Computer vision AI deployable on mobile devices, bank counting machines, and point-of-sale terminals that identifies fake notes through microprint analysis, security thread verification, serial number pattern validation, and UV feature simulation — providing field officers and bank tellers with instant, reliable identification across all denominations.
- **Fraud Network Graph Intelligence —** Graph AI agent that analyses transaction metadata, call records, device fingerprints, and account linkages to map coordinated fraud campaigns — clustering victim reports, scammer infrastructure, and money mule networks into actionable intelligence packages that link across jurisdictions and can be submitted as court-admissible evidence.
- **Geospatial Crime Pattern Intelligence —** Geospatial AI layer for law enforcement that maps fraud complaint locations, counterfeit currency seizure points, and cybercrime hotspots — enabling patrol prioritisation, resource deployment, and inter-district intelligence sharing in near real time through a command centre interface.
- **Citizen Fraud Shield (Multi-channel) —** Conversational AI accessible via WhatsApp, IVR, and mobile app that walks citizens through real-time fraud risk assessment for suspicious calls,

payment requests, or messages — providing instant verdicts, guided reporting to NCRB portals, and advisory in 12 regional languages.

These examples are illustrative only.

SUGGESTED TECHNOLOGIES

- Computer Vision (counterfeit detection, deepfake identification)
- Graph AI & Network Analysis (fraud ring mapping)
- NLP / LLMs (scam script and voice pattern classification)
- Geospatial Intelligence (crime mapping, patrol optimisation)
- Speech AI (voice spoofing and AI-voice detection)
- Agentic AI for multi-source intelligence fusion

EXPECTED DELIVERABLES

- Working Prototype
- Architecture Diagram
- Presentation Deck
- Demo Video

Evaluation Focus Counterfeit detection accuracy across denominations and print quality, digital arrest scam detection precision and recall, fraud network detection lead time before mass victimisation, false positive rate for citizen-facing tools (must be very low), and auditability of intelligence packages for legal admissibility.

JUDGING CRITERIA

Criteria	Weight
Innovation	25%
Business Impact	25%
Technical Excellence	20%
Scalability	15%
User Experience	15%